

Gene	Location	Primer	Direction	Sequence	Reference
<i>COI</i>	mitochondrial	LCO1490 _{a,b}	forward	GGTCAACAAATCATAAAGATATTGG	Folmer et al. (1994)
<i>COI</i>	mitochondrial	HCO2198 _{a,b}	reverse	TAAACTTCAGGGTGACCAAAAAATCA	Folmer et al. (1994)
<i>CAD</i>	nuclear	CD439F _{a,b}	forward	TTCAGTGTACARTTYCAYCCHGARCAYAC	Wild & Maddison (2008)
<i>CAD</i>	nuclear	CD806F _{a,b}	forward	GTNGTNAARATGCCNMGTGGGA	Moulton & Wiegmann (2004)
<i>CAD</i>	nuclear	CD688R _{a,b}	reverse	TGTATACCTAGAGGATCDACRTTYTCCATRTTRCA	Wild & Maddison (2008)
<i>CAD</i>	nuclear	CD1098R2 _{a,b}	reverse	GCTATGTTGTTNGGNAGYTGDCCNCCCAT	Wild & Maddison (2008)
<i>CAD</i>	nuclear	CD1231R _a	reverse	TCCACGTGTTCNANACNGCCATRCA	Wild & Maddison (2008)
<i>Wg</i>	nuclear	wg550F _a	forward	ATGCGTCAGGARTGYAARTGYCAYGGYATGTC	Wild & Maddison (2008)
<i>Wg</i>	nuclear	wg578F _{a,b}	forward	TGCACNGTGAARACYTGCTGGATG	Ward & Downie (2005)
<i>Wg</i>	nuclear	wgAbR _{a,b}	reverse	ACYTCGCAGCACCARTGGAA	Wild & Maddison (2008)
<i>Wg</i>	nuclear	wgAbRZ _a	reverse	CACCTNACYTCRCARCACCARTG	Wild & Maddison (2008)
<i>MSP</i>	nuclear	MSP1F _a	forward	CGAGAYGARGTYGATAARATGATGCA	Maddison and Cooper (2014)
<i>MSP</i>	nuclear	MSP2F _{a,b}	forward	GCTGGACAAAAGGARATYAAYCARTGG	Maddison and Cooper (2014)
<i>MSP</i>	nuclear	MSP1R _{a,b}	reverse	TCWACCAGATCCATCCACTTGACCAT	Maddison and Cooper (2014)
<i>28S</i>	nuclear	D1F _{a,b}	forward	GGGAGGAAAAAGAACTAAC	Ober (2002)
<i>28S</i>	nuclear	D3R _{a,b}	reverse	GCATAGTTCCACCATCTTTC	Ober (2002)
<i>18S</i>	nuclear	18S5 _{a,b}	forward	GACAACCTGGTTGATCCTGCCAGT	Shull et al. (2001)
<i>18S</i>	nuclear	18Sb5 _{a,b}	reverse	TAACCGCAACAACCTTAAT	Shull et al. (2001)
<i>ArgK</i>	nuclear	AK168F _a	forward	CAGGTTTGGARAAYCACGAYTCYGG	Wild & Maddison (2008)
<i>ArgK</i>	nuclear	AK183F _{a,b}	forward	GATTCTGGAGTCGGNATYTAYGCNCCYGAYGC	Wild & Maddison (2008)
<i>ArgK</i>	nuclear	AK939R _{a,b}	reverse	GCCNCCYTCRGYTCRGTGTGYTC	Wild & Maddison (2008)
<i>Topo</i>	nuclear	TP643F _a	forward	GACGATTGGAARTCNAARGARATG	Wild & Maddison (2008)
<i>Topo</i>	nuclear	TP675F _{a,b}	forward	GAGGACCAAGCNGAYACNGTDGGTTGTTG	Wild & Maddison (2008)
<i>Topo</i>	nuclear	TP932R _{a,b}	reverse	GGWCCDGCATCDATDGCCCA	Wild & Maddison (2008)

Table S1. List of primers used in amplification (a) and Sanger sequencing (b).

program name	denaturing temp °C	annealing temp °C	extension temp °C	extension time (s)	final extension time (s)	cycles
C1	94	52	72	45	180	35
C1alt	94	48	72	45	180	35
C2	94	50	72	150	150	35
C3	94	52	72	120	120	37
C4	94	54	72	150	150	35
C5	94	55	72	90	90	37
C6	94	57	72	90	90	31
C7	94	60, 55	72	120	120	9, 30
C8	94	60, 57	72	180	180	9, 27
C9	94	57, 52, 45	72	120	120	6, 6, 36

Table S2. PCR programs. All began with an initial denaturing phase of 180s at 94 °C, with each cycling phase including 30s of denaturing at 94 °C and 30s of annealing. From Maddison (2012).

Gene	Program	Primer 1	Primer 2
18S	C2	18SF5	18Sb5
28S	C1	D1F	D3R
COI	C1alt	LCO1490	HCO2198
CAD2	C7 ¹ , C5 ²	CD439F ^{1,2}	CD688R ² , CD1098R2 ¹
CAD4	C7 ¹ , C5 ²	CD806F ^{1,2}	CD1098R2 ² , CD1231R ¹
wg	C3 ¹ , C4 ²	wg550F ¹ , wg578F ²	wgAbRZ ¹ , wgAbR ²
MSP	C9 ^{1,2}	MSP1F ¹ , MSP2F ¹	MSP1R ^{1,2}
Topo	C9 ^{1,2}	TP643F ¹ , TP675F ²	TP932R ^{1,2}
argK	C8 ¹ , C6 ²	AK168F ¹ , AK183F ²	AK939R ^{1,2}

Table S3. PCR protocols for gene fragments. For nested and hemi-nested reactions, programs and primers for inner and outer reactions are indicated by ¹ and ², respectively.

References

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